

HOPETOUN VIC, Australia

WMO: 948380

Lat: 35.7150S

Lon: 142.3569E

Elev: 78

StdP: 100.39

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.3	1.0	-5.6	2.4	24.3	-3.7	2.8	21.0	12.1	13.5	11.2	13.6	1.5	90	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	39.7	19.1	37.2	18.6	34.9	18.1	21.7	30.5	20.7	30.7	19.8	30.0	6.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.4	23.6	18.1	13.1	23.0	16.5	11.9	22.6	63.8	30.1	59.9	30.5	56.9	30.1	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	10.2	9.1	-2.7	44.0	1.0	1.7	-3.4	45.2	-4.0	46.2	-4.6	47.1	-5.4	48.3		
(5)			-3.1	23.3	1.1	1.4	-3.9	24.3	-4.5	25.1	-5.1	25.8	-5.8	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	24.1	23.1	20.6	16.4	12.4	9.7	9.1	9.9	12.1	15.7	19.1	21.5	(6)
(7)		DBStd	6.39	4.43	4.19	4.24	3.55	2.68	2.11	2.20	2.54	3.23	4.12	4.64	4.60	(7)
(8)		HDD10.0	126	0	0	0	1	7	31	44	33	10	1	0	0	(8)
(9)		HDD18.3	1460	3	7	23	79	183	260	287	262	190	105	45	17	(9)
(10)		CDD10.0	2353	436	366	329	192	82	20	16	29	73	178	273	357	(10)
(11)		CDD18.3	647	181	139	94	21	1	0	0	0	3	24	68	116	(11)
(12)		CDH23.3	9318	2571	1831	1182	302	11	0	0	5	84	490	1075	1768	(12)
(13)		CDH26.7	4841	1480	968	566	98	0	0	0	0	22	207	545	956	(13)
(14)	Wind	WSAvg	4.4	5.0	4.9	4.5	3.9	3.9	3.6	4.1	4.2	4.4	4.7	4.9	5.1	(14)
(15)	Precipitation	PrecAvg	352	26	17	24	28	37	29	33	39	38	36	28	21	(15)
(16)		PrecMax	574	88	89	100	142	94	92	74	77	105	119	113	75	(16)
(17)		PrecMin	104	0	0	0	0	1	3	3	6	5	1	0	1	(17)
(18)		PrecStd	104	25	19	26	29	26	21	19	18	25	26	25	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	41.4	38.6	33.2	24.8	18.7	18.7	23.7	29.5	35.2	38.7	41.9	(19)
MCWB			20.0	19.7	18.9	17.3	14.6	12.9	11.9	12.9	14.4	16.2	18.0	19.1	(20)	
2%		DB	40.4	37.5	35.2	29.3	22.1	16.8	16.5	20.2	25.8	31.8	35.3	38.4	(21)	
		MCWB	19.5	19.2	18.3	16.1	14.0	11.8	11.5	11.8	13.9	15.5	17.3	18.3	(22)	
5%		DB	37.1	34.9	32.5	26.7	19.8	15.5	15.0	17.7	22.3	28.7	32.6	35.1	(23)	
		MCWB	19.0	18.8	17.9	15.1	13.2	11.5	10.9	11.1	12.7	14.5	16.6	17.6	(24)	
10%		DB	34.3	32.6	29.7	24.2	18.1	14.4	13.8	15.8	19.8	25.7	29.8	32.2	(25)	
		MCWB	18.3	18.2	17.2	14.3	12.9	11.0	10.4	10.7	11.9	14.0	16.2	16.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.0	23.3	21.2	18.6	16.2	14.4	13.7	14.0	15.9	18.8	20.7	22.2	(27)
MCDWB			31.2	32.5	31.3	26.6	20.1	16.3	16.0	19.9	23.8	26.6	30.0	29.9	(28)	
2%		WB	21.8	21.6	20.1	17.3	15.1	12.9	12.3	12.7	14.7	17.0	19.4	20.2	(29)	
		MCDWB	30.4	30.5	28.4	24.2	19.9	15.6	15.0	17.9	22.8	25.9	29.0	29.6	(30)	
5%		WB	20.7	20.5	19.1	16.3	14.1	12.1	11.5	11.8	13.6	15.8	18.2	19.1	(31)	
		MCDWB	32.0	30.0	27.9	24.1	18.5	14.5	14.2	16.7	20.7	24.9	28.5	30.6	(32)	
10%		WB	19.6	19.4	18.1	15.3	13.3	11.4	10.7	11.0	12.5	14.7	17.0	17.9	(33)	
		MCDWB	31.7	29.5	28.2	23.1	17.3	13.8	13.2	15.2	18.7	24.1	26.6	30.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	15.6	14.5	13.8	11.5	10.2	10.1	11.9	14.3	16.2	16.0	16.3	(35)
MCDBR			20.2	18.7	17.8	17.1	14.0	11.6	11.5	15.4	18.4	21.0	19.8	19.9	(36)	
MCWBR			6.6	6.1	6.4	7.8	7.6	7.4	7.1	8.2	8.6	8.5	6.8	7.0	(37)	
5% WB		MCDWB	15.5	14.2	14.2	13.8	11.4	9.9	10.0	13.0	15.6	16.4	15.9	15.1	(38)	
		MCWBR	6.4	5.2	6.0	6.7	6.5	6.8	6.8	7.7	8.0	7.5	6.3	6.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.340	0.335	0.318	0.308	0.300	0.290	0.290	0.301	0.322	0.332	0.340	0.331	(40)	
taud		2.481	2.508	2.548	2.553	2.556	2.581	2.581	2.529	2.464	2.447	2.456	2.495	(41)		
Ebn at Noon		995	978	958	910	860	844	865	903	937	970	988	1008	(42)		
Edh at Noon		116	108	96	84	74	67	71	85	103	114	118	115	(43)		
(44)	All-Sky Solar Radiation	RadAvg	7.84	7.01	5.67	4.12	2.83	2.27	2.44	3.40	4.66	6.13	7.03	7.77	(44)	
(45)		RadStd	0.35	0.34	0.26	0.19	0.13	0.16	0.12	0.21	0.34	0.39	0.40	0.31	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+122	+87	

Nomenclature: See separate page